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EE-206

2-stage OTA Design in TSMC 180nm technology

$$V_{Tn} = 0.37V; V_{Tp} = 0.39V; \mu_n C_{ox} = 230 \mu A/V^2; \\ \mu_p C_{ox} = 100 \mu A/V^2; V_{dd} = 1.8V; L_{min} = 0.18 \mu m; W_{min} = 0.27 \mu m$$

Specifications :- i.) DC Gain $\sim 40dB$
ii.) Op-amp to be used for non-inverting amplifier of gain = 2.

Assumptions Taken :- i.) Max Power Consumption = $500 \mu W$
ii.) Phase Margin $\sim 60^\circ$
iii.) Gate Overdrive Voltage = $200mV$
~~iv.)~~

$$V_{dd}(I_1 + I_2) \leq 500 \mu W \\ I_1 + I_2 \leq \frac{500 \times 10^{-6}}{1.8}$$

$$\begin{cases} I_1 \rightarrow \text{Stage 1 Tail Current} \\ I_2 \rightarrow \text{Stage 2 Current} \end{cases}$$

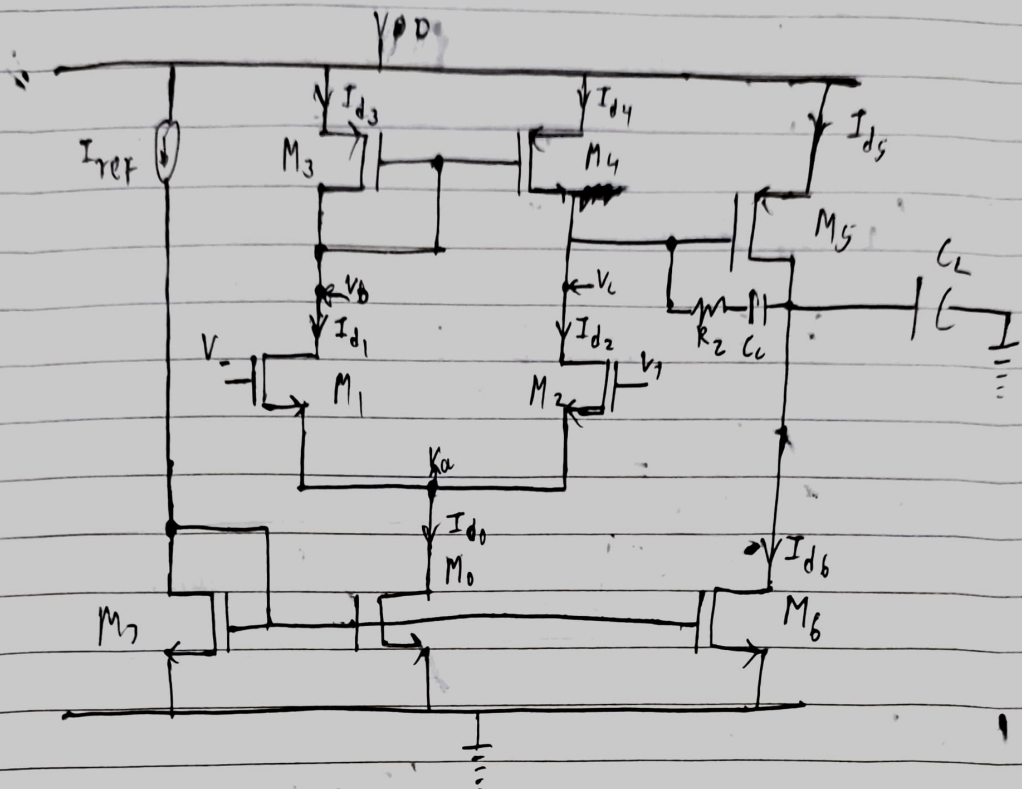
$$I_1 + I_2 \leq 277.77 \mu A$$

$$\text{Let } I_1 = I_2 = 100 \mu A$$

$$iv.) I_{ref} = 100 \left(\frac{1}{10} \right) = 10 \mu A$$

$$v.) \lambda = 0.1 V^{-1}$$

Circuit Diagram :-



Calculations :-

1) M_7 :- (NMOS)

$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_7 (\Delta V)^2$$

$$\left(\begin{array}{l} I_D = 10 \mu A, \Delta V = 0.2 V \\ \mu_n C_{ox} = 230 \mu A/V^2 \end{array} \right)$$

$$10^{-5} = \frac{1}{2} 230 \times 10^{-6} \left(\frac{W}{L}\right)_7 (0.2)^2$$

$$\left(\frac{W}{L}\right)_7 = 2.17$$

$$\text{Take } \left(\frac{W}{L}\right)_7 = 2.5 \quad \{L = 1 \mu m, W = 2.5 \mu m\}$$

2) M_0 :- (NMOS)

$$10^{-4} = \frac{1}{2} 230 \times 10^{-6} \left(\frac{W}{L}\right)_0 (0.2)^2$$

$$(I_D = 100 \mu A)$$

(L has been chosen to be uniform ~~or~~ to keep λ uniform as $\lambda \propto \frac{1}{L}$)

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$$\left(\frac{W}{L}\right)_0 = 21.7$$

$$\text{Take } \left(\frac{W}{L}\right)_0 = 25 \quad (L = 1\mu\text{m}, W = 25\mu\text{m})$$

3.) M_1, M_2 (NMOS)

$$5 \times 10^{-5} = \frac{1}{2} \cdot 230 \times 10^{-6} \left(\frac{W}{L}\right)_{1,2} (0.2)^2 \quad (I_D = 50\mu\text{A})$$

$$\left(\frac{W}{L}\right)_{1,2} = 10.87$$

$$\text{Take } \left(\frac{W}{L}\right)_{1,2} = 15 \quad (L = 1\mu\text{m}, W = 15\mu\text{m})$$

4.) M_3, M_4 (PMOS)

$$5 \times 10^{-5} = \frac{1}{2} \cdot 100 \times 10^{-6} \left(\frac{W}{L}\right)_{3,4} (0.2)^2 \quad (I_D = 50\mu\text{A}, \mu_{p\text{ox}} = 100\mu\text{A/V}^2)$$

$$\left(\frac{W}{L}\right)_{3,4} = 25 \quad (L = 1\mu\text{m}, W = 25\mu\text{m})$$

5.) M_5 (PMOS)

$$(I_D = 100\mu\text{A})$$

$$10^{-4} = \frac{1}{2} \cdot 100 \times 10^{-6} \left(\frac{W}{L}\right)_5 (0.2)^2$$

$$\left(\frac{W}{L}\right)_5 = 50 \quad (L = 1\mu\text{m}, W = 50\mu\text{m})$$

6.) M_6 (NMOS)

$$(I_D = 100\mu\text{A})$$

$$10^{-4} = \frac{1}{2} \cdot 230 \times 10^{-6} \left(\frac{W}{L}\right)_6 (0.2)^2$$

$$\left(\frac{W}{L}\right)_6 = 21.7$$

$$\text{Take } \left(\frac{W}{L}\right)_6 = 25 \quad (L = 1 \mu\text{m}, W = 25 \mu\text{m})$$

Use $g_m = \frac{2I_D}{\Delta V}$ for following
Zero compensation Resistor theoretical table

Mosfet No.	Type	I_D	g_m	$\left(\frac{W}{L}\right)$	W	L
M_0	NMOS	$100 \mu\text{A}$	$1000 \mu\text{S}$	25	$25 \mu\text{m}$	$1 \mu\text{m}$
$M_{1,2}$	NMOS	$50 \mu\text{A}$	$500 \mu\text{S}$	25	$25 \mu\text{m}$	$1 \mu\text{m}$
$M_{3,4}$	PMOS	$50 \mu\text{A}$	$500 \mu\text{S}$	25	$25 \mu\text{m}$	$1 \mu\text{m}$
M_5	PMOS	$100 \mu\text{A}$	$1000 \mu\text{S}$	50	$50 \mu\text{m}$	$1 \mu\text{m}$
M_6	NMOS	$100 \mu\text{A}$	$1000 \mu\text{S}$	25	$25 \mu\text{m}$	$1 \mu\text{m}$
M_7	NMOS	$10 \mu\text{A}$	$100 \mu\text{S}$	2.5	$25 \mu\text{m}$	$1 \mu\text{m}$

Zero compensation Resistor (R_2)

$$R_2 = \frac{1}{g_{m5}} = \frac{1}{1000 \times 10^{-6}} = 1 \text{ k}\Omega$$

Pole compensation

$$PM = 60^\circ$$

$$\therefore \geq 180^\circ - 90^\circ - \tan^{-1}\left(\frac{W_u}{W_o}\right) = 60^\circ$$

$$\tan^{-1}\left(\frac{W_u}{W_o}\right) = 30^\circ$$

$$\frac{W_u}{W_o} = \frac{1}{\sqrt{3}}$$

$$W_u = \frac{g_{m1}}{C_C}$$

$$W_o = \frac{g_{m5}}{C_L}$$

$$\frac{500}{C_C} \cdot \frac{C_L}{1000} = \frac{1}{\sqrt{3}}$$

$$C_c = 0.86 C_L$$

This would cause too small of a bandwidth as C_c ends up being too close to C_L so we will take $C_c = 0.5 C_L$

$$\text{Take } C_L = 10 \text{ pF}$$

$$C_c = 5 \text{ pF}$$

Input Swing Limits:-

For $V_{in, \min}$:- M_1 in triode

$$V_{DS1} = V_{S1}$$

$$V_{GS1} = V_{GS1} - V_{S1}$$

$$\Delta V + V_{Thn} = V_{in} - V_{S1}$$

$$V_{S1} = V_{in} - (0.2 + 0.37)$$

$$V_{S1} = V_{in} - (0.57)$$

$$V_{DS1} \geq \Delta V$$

$$V_{S1} \geq 0.2 \text{ V}$$

$$V_{in} - 0.57 \geq 0.2$$

$$V_{in, \min} = 0.77 \text{ V}$$

For $V_{in, \max}$:- $M_{1,2}$ in triode:-

$$V_{D1} = V_{DD} - |V_{GS3}|$$

$$= 1.8 - (0.2 + 0.37)$$

$$= 1.8 - 0.57$$

$$V_{D1} = 1.23 \text{ V}$$

$$V_{DS1} \geq \Delta V$$

$$1.23 - (V_{in} - 0.57) \geq 0.2$$

$$1.8 - V_{in} \geq 0.2$$

$$V_{in} \leq 1.6V$$

$$V_{in, \text{max}} = 1.6V$$

Input Common Mode Range (ICMR) = $0.77V$ to $1.6V$
Range = $0.83V$

Output Swing Limits

$V_{out, \text{min}}$ if M_2 in triode:-

$$V_{out} \geq \Delta V$$

$$V_{out, \text{min}} = 0.2V$$

$V_{out, \text{max}}$ if M_1 in triode:-

$$V_{out} \leq V_{DD} - \Delta V$$

$$V_{out, \text{max}} = 1.8 - 0.2$$

$$V_{out, \text{max}} = 1.6V$$

To ensure maximum output swing, the output must be biased at $\frac{(0.2 + 1.6)}{2} = 0.9V$.

$$r_o = \frac{1}{\lambda I_D}$$

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Operating Point Simulation:-

MOSFET No.	$\Delta V(V_{ov})$	I_D	g_m	MOSFET Type	$r_o = \frac{1}{\lambda I_D}$
M ₀	0.221V	94.76 μA	857.55 μS	NMOS	105 k Ω
M _{1, M_2}	0.27V	47.38 μA	350.96 μS	NMOS	211 k Ω
M _{3, M_4}	0.28V	47.38 μA	338.42 μS	PMOS	211 k Ω
M ₅	0.28V	98.16 μA	701.14 μS	PMOS	102 k Ω
M ₆	0.221V	98.16 μA	888.32 μS	NMOS	102 k Ω
M ₇	0.221V	10 μA	90.49 μS	NMOS	1 M Ω

~~DC Gain~~

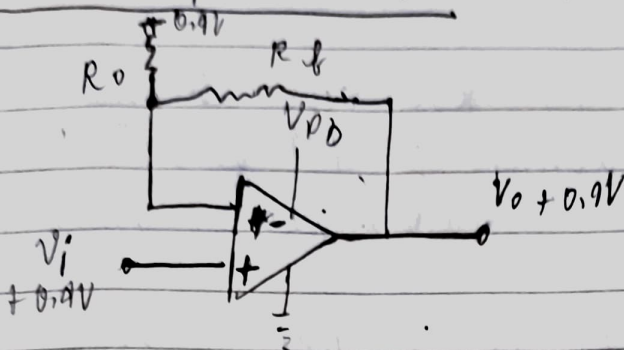
$$\text{Power Consumption} = V_{DD} \times (I_{M_0} + I_{M_6})$$

$$= 1.8 \times (94.76 + 98.16)$$

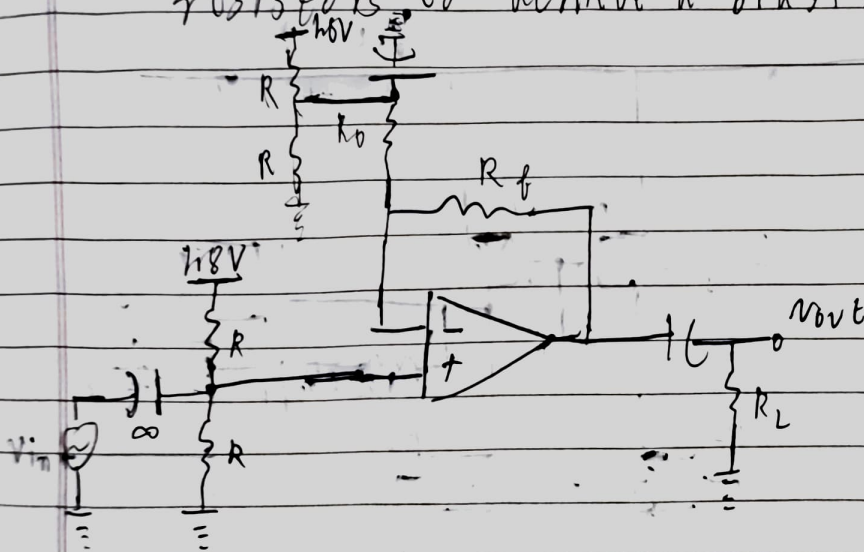
$$P = 34.7.256 \mu W$$

DC Gain (Obtained From ~~Transient~~ ^{Frequency} Response) $\approx 70 \text{ dB}$ (Greater than 40 dB)

Closed loop gain of 2:-



A C signal must be biased at 0.9V to achieve maximum output swing limits. For this, we can use the 1.8V V_{DD} and voltage dividers using resistors to achieve a biasing of 0.9V:-



Take $R_f = 100k\Omega$

$$\text{Gain} = 1 + \frac{R_f}{R_o}$$

$$2 = 1 + \frac{R_f}{R_o}$$

$$R_f = R_o$$

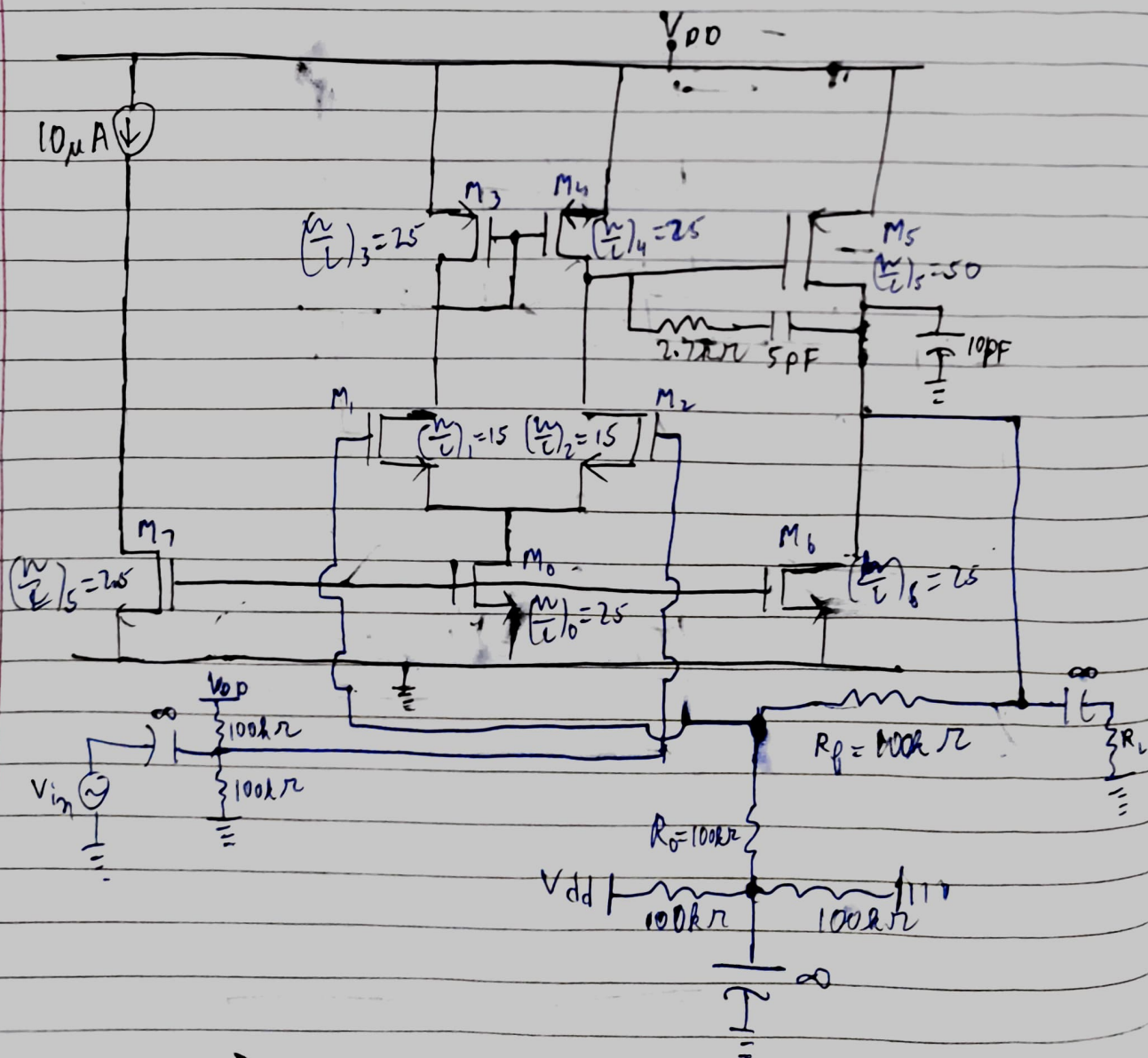
Take $R_f = R_o = 100k\Omega$

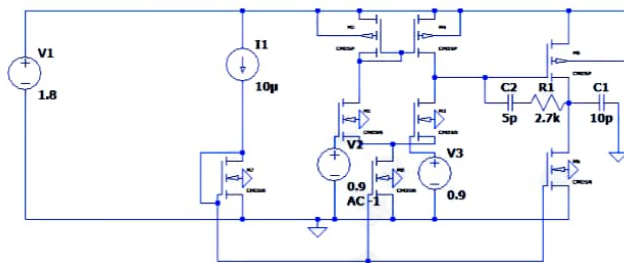
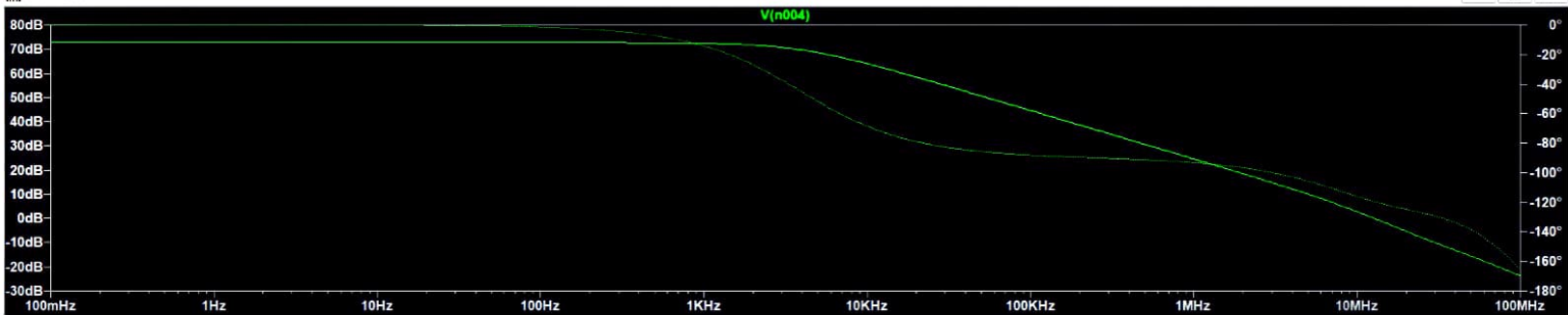
V_{ov} and I_D are close to assumed values.

PM with $R_2 = 1k\Omega$ gives a small PM of 30° so to achieve a PM $\sim 60^\circ$, I tried out multiple R_2 values and chose $R_{2\text{final}} = 2.7k\Omega$.

I chose not to change C_c values to conserve bandwidth.

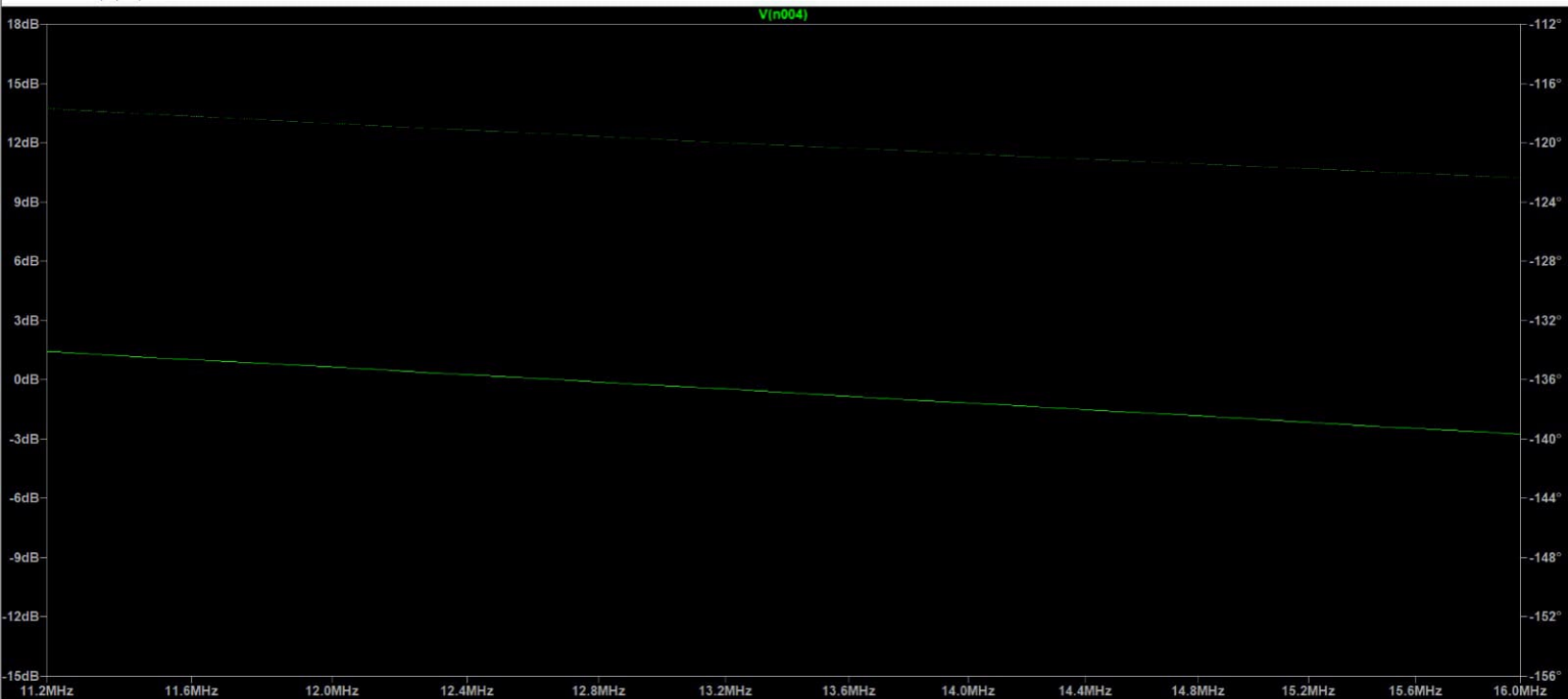
Final Ckt. Diagram



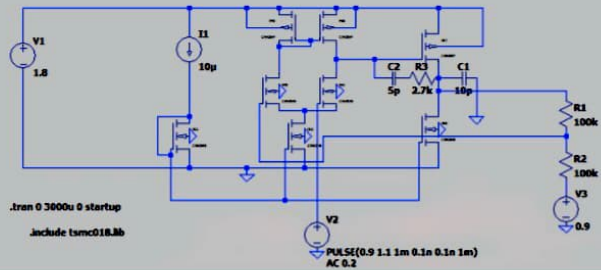


```
.ac dec 100 0.1 100Meg
```

```
.include tsmc018.lib
```



x = 12.730MHz y = 12.376dB, -119.499°





--- Operating Point ---

V(n006):	0.260781	voltage
V(n002):	1.12818	voltage
V(n005):	0.9	voltage
V(n008):	0.9	voltage
V(n007):	0.591155	voltage
V(n001):	1.8	voltage
V(n003):	1.12818	voltage
V(n004):	0.539495	voltage
V(p001):	0.539495	voltage
Id(M7):	1e-05	device_current
Ig(M7):	0	device_current
Is(M7):	-1e-05	device_current
Ib(M7):	-6.01155e-13	device_current
I(V1):	-0.000202924	device_current
Id(M2):	4.73828e-05	device_current
Ig(M2):	0	device_current
Is(M2):	-4.73828e-05	device_current
Ib(M2):	-1.40896e-12	device_current
Id(M3):	4.73828e-05	device_current
Ig(M3):	-0	device_current
Is(M3):	-4.73828e-05	device_current
Ib(M3):	6.81822e-13	device_current
I(V2):	0	device_current
I(V3):	0	device_current
I(C1):	5.39495e-24	device_current
I(C2):	-2.94341e-24	device_current
Id(M0):	9.47657e-05	device_current
Ig(M0):	0	device_current

Id(M0) :	9.47657e-05	device_current
Ig(M0) :	0	device_current
Is(M0) :	-9.47657e-05	device_current
Ib(M0) :	-2.70781e-13	device_current
Id(M1) :	4.73828e-05	device_current
Ig(M1) :	0	device_current
Is(M1) :	-4.73828e-05	device_current
Ib(M1) :	-1.40896e-12	device_current
Id(M4) :	4.73828e-05	device_current
Ig(M4) :	-0	device_current
Is(M4) :	-4.73828e-05	device_current
Ib(M4) :	6.81822e-13	device_current
Id(M5) :	9.81586e-05	device_current
Ig(M5) :	-0	device_current
Is(M5) :	-9.81586e-05	device_current
Ib(M5) :	1.2705e-12	device_current
Id(M6) :	9.81586e-05	device_current
Ig(M6) :	0	device_current
Is(M6) :	-9.81586e-05	device_current
Ib(M6) :	-5.49495e-13	device_current
I(I1) :	1e-05	device_current
I(R1) :	0	device_current

Final Simulation Results:-

Open Loop Gain $\sim 70 \text{ dB}$ ~~+72.68~~ (72.68 dB exactly)

Closed Loop Gain ~ 2

Bandwidth $\Rightarrow$ Frequency till which open loop gain drops by 3 dB
~~at~~ i.e. freq. at which gain reaches 69.68 dB.

Bandwidth $= 4.7 \text{ kHz}$

Phase Margin $\sim 60^\circ$

Power Consumption $\sim 350 \mu \text{W}$

Unity Gain (0 dB) Frequency $\sim 12.5 \text{ MHz}$